

Novel anti-microbial therapies for dental plaque-related diseases.

Control of dental plaque-related diseases has traditionally relied on non-specific removal of plaque by mechanical means. As our knowledge of oral disease mechanisms increases, future treatment is likely to be more targeted, for example at small groups of organisms, single species or at key virulence factors they produce. The aim of this review is to consider the current status as regards novel treatment approaches. Maintenance of oral hygiene often includes use of chemical agents; however, increasing problems of resistance to synthetic antimicrobials have encouraged the search for alternative natural products. Plants are the source of more than 25% of prescription and over-the-counter preparations, and the potential of natural agents for oral prophylaxis will therefore be considered. Targeted approaches may be directed at the black-pigmented anaerobes associated with periodontitis. Such pigments provide an opportunity for targeted phototherapy with high-intensity monochromatic light. Studies to date have demonstrated selective killing of *Porphyromonas gingivalis* and *Prevotella intermedia* in biofilms. Functional inhibition approaches, including the use of protease inhibitors, are also being explored to control periodontitis. Replacement therapy by which a resident pathogen is replaced with a non-pathogenic bacteriocin-producing variant is currently under development with respect to *Streptococcus mutans* and dental caries.